

Notes: Khanna and Palepu (JF, 2000)
**Is Group Affiliation Profitable in Emerging Markets? An Analysis of
Diversified Indian Business Groups**

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Sample construction	4
3.2	Group affiliation and group structure	4
3.3	Performance measures	4
3.4	Diversification measures	5
3.5	Ownership measures	5
4	Models and empirical specifications	5
4.1	Baseline performance regression	5
4.2	Diversification-category specification	5
4.3	Quadratic diversification specification	6
4.4	Group-level robustness	6
4.5	Mechanism tests	6
5	Identification and empirical design	6
5.1	What identifies the main patterns	6
5.2	Why India is a useful empirical setting	6
5.3	Main threats to identification	7
5.4	Identification takeaway	7
6	Section-by-section study guide	7
7	Tables and figures	8
7.1	Table I: Descriptive Statistics + Table II: Diversification Measures	8
7.2	Table III: Tobin's q and ROA by Group Diversification and Group Size Categories + Table IV: The Effects of Group Diversification and Group Size, Part 1	9
7.3	Table IV: The Effects of Group Diversification and Group Size, Part 2 + Table A1: The Institutional Context in India	10
8	Detailed result map	11
8.1	Result 1: No simple average group premium	11
8.2	Result 2: Nonmonotonic group-diversification effect	11
8.3	Result 3: Group size alone is not the main explanation	11

8.4	Result 4: Mechanism evidence favors institutional intermediation	12
8.5	Result 5: Institutional context matters	12
9	Short summary	12
10	Conclusion	12
10.1	Core conclusion	12
10.2	Economic intuition	13
10.3	Why the result is incremental	13
10.4	Policy implication	13
10.5	Limitations	13
10.6	Takeaway	13

1 Big picture

- **Question.** Are firms affiliated with diversified business groups in an emerging market more or less profitable than unaffiliated firms?
- **Setting.** The paper studies Indian private-sector firms listed on the Bombay Stock Exchange in 1993. India is useful because it has many business groups, good firm-level public data, separate legal entities within groups, and clear group classifications from CMIE.
- **Motivation.** In advanced economies, conglomerate diversification is often associated with value loss. In emerging markets, however, markets for capital, labor, product quality, governance, and information are incomplete. Business groups might internalize missing institutions.
- **Core trade-off.** Group affiliation can create value by providing internal intermediation, reputation, managerial talent, political access, and capital-market access. It can also destroy value through agency problems, tunneling, inefficient cross-subsidization, weak minority protection, and over-complex coordination.
- **Main result.** The relation between performance and group diversification is nonmonotonic. Firms in intermediate-diversification groups underperform, while affiliates of the most diversified groups outperform unaffiliated firms on Tobin's q and do not underperform on ROA.
- **Important distinction.** The paper is not measuring U.S.-style conglomerate divisions. Indian group affiliates are legally separate listed firms, often with distinct minority shareholders, but tied by family ownership, boards, reputation, and coordination.
- **Mechanism evidence.** The authors find that group affiliates, especially in large diversified groups, have better access to foreign capital markets. They find less support for internal capital markets and no special evidence that concentrated insider ownership uniquely drives group performance.
- **Interpretive thesis.** The largest diversified Indian groups may replicate missing market institutions. Smaller or less diversified groups may not have the scale, managerial systems, or political connections to create benefits, while still bearing agency costs.
- **Takeaway.** In emerging markets, diversification through business groups can be valuable at sufficient scale and scope. The institutional context reverses the standard advanced-market presumption that diversification necessarily destroys value.

2 Incremental contribution of the paper

- **Emerging-market theory of business groups.** The paper moves the business-group debate beyond a simple “diversification discount” view. It argues that in countries with institutional voids, diversified groups may substitute for missing financial, labor, product, and regulatory intermediaries.
- **Context-dependent view of diversification.** The study shows that the valuation effects of diversification depend on whether external markets are well developed. The same organizational form that is costly in the United States may be valuable in India.
- **Firm-level evidence within groups.** Instead of treating the group as a single conglomerate, the paper studies listed affiliate firms. This matters because each affiliate has public shareholders and its own performance measure.
- **Nonmonotonicity.** A central contribution is documenting a U-shaped relation between group diversification and affiliate performance: initial diversification is associated with lower performance, but very high diversification is associated with higher performance.
- **Institutional-void mechanism.** The paper formalizes the idea that large groups may create internal substitutes for financial analysts, investment banks, venture capitalists, managerial labor markets, product-market reputation, and government-interface capacity.

- **Distinguishing size from diversification.** The regressions control separately for group size and group diversification. This matters because the mechanism is not simply “big groups are better”; the strongest results are tied to breadth of industries rather than the number of firms.
- **Mechanism tests.** The paper examines international capital access, joint ventures, insider ownership, and internal capital markets. This broadens the contribution from a valuation pattern to a richer institutional explanation.
- **Caution about causality.** The paper is explicit that its evidence is cross-sectional and cannot fully eliminate endogeneity. Its incremental contribution is best read as a foundational empirical diagnosis of why groups may matter in emerging markets.

3 Data and measurement

3.1 Sample construction

- The data come primarily from the Centre for Monitoring the Indian Economy (CMIE).
- The sample includes all Indian private-sector firms listed on the Bombay Stock Exchange in 1993 with the required financial and stock-price information.
- The final sample contains 1,309 firms: 655 group-affiliated firms and 654 unaffiliated firms.
- Group affiliation follows the CMIE classification, which uses corporate histories, filings, public announcements, government reports, and business-group records.
- The authors focus on 1993 because it is the year with the most complete data coverage.

Interpretation: The sample design allows the authors to compare legally separate listed firms that differ in group affiliation. It also avoids the problem of inferring divisional performance inside a single conglomerate.

3.2 Group affiliation and group structure

- Indian business groups are collections of legally separate firms tied by common family ownership, board links, reputation, and coordination.
- Unlike Japanese keiretsu, Indian business groups did not have group-specific banks because banking was nationalized and regulations restricted ownership patterns.
- Unlike U.S. conglomerates, Indian business groups do not consolidate many divisions into one publicly traded corporation.
- The firm-level structure limits some forms of internal capital-market abuse but does not eliminate agency problems.

Interpretation: The institutional form is hybrid: less integrated than a U.S. conglomerate, but more coordinated than a set of unrelated firms.

3.3 Performance measures

The paper uses two main performance measures:

$$Tobin's\ q_i = \frac{\text{market value of equity}_i + \text{book value of preferred stock}_i + \text{book value of debt}_i}{\text{book value of assets}_i},$$

$$ROA_i = \frac{\text{net income}_i + (1 - \tau_i) \times \text{interest}_i}{\text{total assets}_i}.$$

Interpretation: Tobin’s q is forward-looking and market-based, while ROA is accounting-based. Similar patterns across both measures make the findings more robust.

3.4 Diversification measures

- **INDCNT:** the number of two-digit SIC-equivalent industries represented in the group.
- **ENTROPY:** a Palepu-style entropy measure that combines related and unrelated diversification.
- **CONCENTRIC:** a Caves et al.-style measure that captures the degree of relatedness of diversification.
- **HERFINDAHL:** the sum of squared industry sales shares, where higher values mean more concentration and therefore less diversification.

Interpretation: Because India did not have a standard SIC classification, the authors construct an SIC-equivalent classification from firms’ product descriptions. This is a major measurement step.

3.5 Ownership measures

- Domestic institutional ownership.
- Foreign institutional ownership.
- Insider ownership, including family ownership and stakes held through other group companies.
- Directors’ ownership.
- Top-50 ownership excluding the above categories.

Interpretation: Ownership variables are used both to compare group and nongroup firms and to test whether monitoring or entrenchment explains performance differences.

4 Models and empirical specifications

4.1 Baseline performance regression

The baseline firm-level specification can be written as:

$$Performance_i = \alpha + \beta_1 Group_i + \gamma' X_i + \delta_{industry(i)} + \varepsilon_i,$$

where $Performance_i$ is either Tobin’s q or ROA, $Group_i$ indicates diversified-group membership, X_i includes firm controls such as log sales and age, and $\delta_{industry(i)}$ are industry dummies.

Interpretation: The coefficient on $Group_i$ captures the average difference between group-affiliated and unaffiliated firms after controlling for firm size, age, and industry.

4.2 Diversification-category specification

$$Performance_i = \alpha + \beta_L Least_i + \beta_M Intermediate_i + \beta_H Most_i + \gamma' X_i + \delta_{industry(i)} + \varepsilon_i.$$

Interpretation: This specification allows group affiliation to have different effects depending on the diversification breadth of the group. It is the simplest way to test the paper’s core U-shape.

4.3 Quadratic diversification specification

$$Performance_i = \alpha + \beta_1 INDCNT_i + \beta_2 INDCNT_i^2 + \beta_3 GroupSize_i + \gamma' X_i + \delta_{industry(i)} + \varepsilon_i.$$

Interpretation: A negative β_1 and positive β_2 imply that performance first falls and then rises with group diversification. The turning point is approximately $-\beta_1/(2\beta_2)$.

4.4 Group-level robustness

The group-level performance measure is constructed as:

$$IndustryAdjustedGroupQ_g = GroupQ_g - \sum_j w_{gj} IndustryQ_j,$$

where w_{gj} is the share of group assets in industry j .

Interpretation: This asks whether a group as a portfolio of affiliates outperforms a comparable portfolio of unaffiliated focused firms in the same industries.

4.5 Mechanism tests

- **International capital access:** group affiliates' access to foreign capital, global investors, or joint ventures.
- **Ownership/monitoring:** whether insider ownership affects performance differently in group and non-group firms.
- **Internal capital markets:** whether investment is less sensitive to cash flow inside groups.

Interpretation: These tests ask whether the performance pattern is connected to plausible benefits or costs of group affiliation.

5 Identification and empirical design

5.1 What identifies the main patterns

- The main comparisons are cross-sectional comparisons between group-affiliated and unaffiliated firms in 1993.
- Industry dummies control for industry-level valuation and profitability differences.
- Firm controls absorb observable differences in scale and age.
- Diversification measures exploit cross-sectional differences across group structures.

Interpretation: The design identifies conditional correlations, not randomized causal effects.

5.2 Why India is a useful empirical setting

- India had many diversified business groups, generating meaningful variation in group structure.
- Group affiliates were typically publicly listed, providing firm-level market and accounting data.

- The institutional environment had capital, labor, product, and regulatory market imperfections.
- Group membership was not typically chosen through active acquisitions during the period, reducing one endogeneity concern.

Interpretation: The setting is ideal for testing whether groups substitute for missing institutions, though it remains cross-sectional.

5.3 Main threats to identification

- **Endogeneity of group membership:** more profitable firms may be sorted into large groups or groups may grow around successful firms.
- **Measurement error:** diversification measures depend on constructed SIC-equivalent classifications.
- **Accounting distortions:** Indian accounting treatment of intercorporate investments can affect Tobin's q and ROA in different directions.
- **Political connections:** the most diversified groups may benefit from political access rather than purely institutional intermediation.
- **Unobserved family quality:** group families may differ in managerial ability or reputation.

Interpretation: The authors acknowledge these limitations and interpret the findings as evidence consistent with institutional intermediation, not as definitive causal proof.

5.4 Identification takeaway

The paper is strongest as a foundational empirical comparison showing that business-group diversification has a different performance relation in emerging markets than in the United States. The most important empirical pattern is not simply a group premium or discount, but the nonmonotonic relation between group diversification and affiliate performance.

6 Section-by-section study guide

- **Introduction.** Frames the theoretical ambiguity: groups can substitute for missing markets or worsen agency problems.
- **Data and institutional details.** Explains why India is useful and defines the sample, group classification, and performance measures.
- **Relative performance.** Presents univariate and multivariate evidence of the U-shaped relation between group diversification and performance.
- **Sources of costs and benefits.** Examines mechanisms: international access, ownership, and internal capital markets.
- **Conclusion.** Interprets the evidence as showing that the largest diversified groups may replicate missing institutions, while most groups do not.

7 Tables and figures

7.1 Table I: Descriptive Statistics + Table II: Diversification Measures

Read:

- Table I shows that group affiliates are much larger and older than nongroup firms: mean sales are 848 million rupees for group affiliates versus 209 million for nongroup firms.
- Group affiliates have higher domestic institutional, foreign institutional, and insider ownership.
- Average ROA and Tobin's q are similar across group and nongroup firms, so the unconditional group-affiliation comparison does not show a large average premium.
- Table II defines the main diversification measures: number of industries, entropy, concentric diversification, and Herfindahl concentration.
- The average group affiliate belongs to a group operating in 3.76 industries; the median is 3 industries.
- Diversification measures are strongly correlated, with INDCNT positively correlated with entropy and negatively correlated with Herfindahl concentration.

Detailed interpretation: Table I is important because group affiliates and nongroup firms are not otherwise similar. The group firms are larger, older, and more institutionally owned. These differences justify the multivariate regressions that control for firm characteristics and industry effects. Table II is important because the entire empirical argument depends on measuring group diversification. The paper uses several alternatives to avoid relying only on a simple count.

Economic message: The raw data show that group affiliation is associated with different firm characteristics, not simply different performance. The study's central result appears only after looking inside the category of group affiliates by diversification intensity.

Table I
Descriptive Statistics

The table reports descriptive statistics of firms affiliated with diversified Indian business group relative to those firms that are not thus affiliated. The data consist of all Indian private sector firms that are publicly traded on the Bombay Stock Exchange in 1993, and are obtained from the Centre for Monitoring the Indian Economy (CMIE). Tobin's q is approximated by (market value of equity + book value of preferred stock + book value of debt)/(book value of assets). ROA is defined as (net income + (1 - tax rate) * interest)/(total assets), where an average tax rate is calculated for each firm. Age reflects the number of years since incorporation. Sales are in rupees millions (\$1 = Rs. 30). Number of industries is the simplest of several measures of group diversification that we explore. It is obtained by first constructing an SIC equivalent classification for Indian firms, since no such classification exists, then assigning two-digit SIC category to each firm, and finally creating the diversification measure for a group as a function of the primary industries of each of its member firms. Domestic institutional ownership measures the ownership stake of all state-run financial institutions. Foreign institutional ownership includes ownership by foreign corporations as well as by foreign financial intermediaries. Insider ownership includes the stake held by the family associated with the firm (for groups, this stake may be held through other group firms). Directors' ownership includes the stakes held by all nonfamily directors. Top-50 ownership includes the stakes of the 50 largest individual shareholders in the firm, other than those that fall in the above categories. The missing category is dispersed ownership. The first two columns report means for group affiliates and nongroup firms; significance levels are based on two-tailed difference means t -tests. The last two columns report medians of group affiliates and nongroup firm significance levels are based on Wilcoxon signed-rank tests of the difference of medians.

	Mean		Median	
	Group Affiliates	Nongroup Firms	Group Affiliates	Nongroup Firms
Panel A: Summary Statistics				
Number of industries	655	654	655	654
Sales (in rupees millions)	848.20	208.52***	400.95	114.02***
Assets (in rupees millions)	1,035.75	281.02***	415.34	130.29***
Age (years)	24.11	15.15***	17.00	10.00***
ROA	8.68%	8.38%	9.05%	8.90%
Tobin's q	1.40	1.38	1.14	1.07**
Panel B: Ownership Structure				
Domestic institutional ownership	15.60%	11.30%**	13.30%	6.50%* [†]
Foreign institutional ownership	10.10%	7.40%***	2.30%	0.90%* [†]
Insider ownership	31.90%	20.80%***	31.30%	17.10%* [†]
Directors' ownership	5.70%	14.20%**	1.10%	10.70%* [†]
Top-50 ownership (excluding above)	4.90%	7.60%**	3.20%	5.00%* [†]

*** and ** denote significance at the 1 and 5 percent levels, respectively.

Table I: Descriptive Statistics

Table II
Diversification Measures

This table reports several measures of group diversification. Since there is no numeric classification of industries in India, we first create an SIC-equivalent classification system for India firms. Starting with a description of each firm's product range provided by the Centre for Monitoring the Indian Economy (CMIE), we first assign a dominant two-digit SIC category that most closely represents that firm's primary area of activity. Then for each group we construct measures of group diversification as a function of the primary areas of activity of all the firms affiliated with that group. The simplest measure, INDCNT, is a count of industries represented within the group. ENTROPY is calculated as the sum of an entropy index of related diversification and an entropy index of unrelated diversification following Palepu (1985). The index of related diversification is based on the various firms within a group that operate within the same two-digit SIC equivalent industry, and is calculated as a function of all such instances within the group. The index of unrelated diversification is a function of different firms within the group operating in different two-digit SIC equivalent industries. Each index is a weighted average of the ratio of firm sales to group sales in that two-digit category (for related diversification) or a weighted average of the ratio of sales within a particular two-digit industry to total group sales (for unrelated diversification), with the weights given by the logarithm of the reciprocal of the ratio. HERFINDAHL is calculated as the sum of the squares of each industry sales as a proportion of total group sales. Sales in an industry is the sum of sales in the industry for all group firms in that industry. CONCENTRIC is a weighted average of each firm's share of group sales, following Caves et al. (1980). The weight includes the contribution of the shares of group sales of all other firms in the group, and accounts for whether or not the firms are in the same two-digit SIC equivalent industry. Panel A reports summary statistics for each measure of diversification for the sample of group affiliates, and a column separately giving the value that the diversification measure takes for unaffiliated firms. Panel B reports pairwise Pearson correlation coefficients for the group diversification measures.

Panel A: Summary Statistics					
Diversification Measure	Group Affiliates				Nongroup Firms Value
	Mean	Median	Min	Max	
Number of different industries in group: INDCNT	3.76	3.00	1.00	13.00	1.00
Total entropy measure of diversification: ENTROPY	1.20	1.20	0.00	2.37	0.00
Concentric measure of diversification: CONCENTRIC	0.92	0.95	0.00	1.71	0.00
Herfindahl measure of diversification: HERFINDAHL	0.62	0.57	0.15	1.00	1.00
Panel B: Pearson's Correlation Coefficients for Diversification Measures for All Group Affiliates***					
	INDCNT	ENTROPY	CONCENTRIC	HERFINDAHL	
INDCNT	1.00				
ENTROPY	0.75	1.00			
CONCENTRIC	0.72	0.97	1.00		
HERFINDAHL	-0.75	-0.83	-0.93	1.00	

Table II: Diversification Measures

7.2 Table III: Tobin's q and ROA by Group Diversification and Group Size Categories + Table IV: The Effects of Group Diversification and Group Size, Part 1

Read:

- Table III shows the main nonmonotonic pattern. Least-diversified group affiliates look similar to nongroup firms, intermediate-diversified affiliates underperform, and most-diversified affiliates outperform on Tobin's q .
- The most diversified groups have mean Tobin's q of 1.655 and median Tobin's q of 1.393, both economically high relative to nongroup firms.
- Intermediate-diversified groups have mean Tobin's q of 1.255 and median ROA of 0.077, indicating underperformance.
- Group-size categories show a similar but weaker pattern: the largest groups look better than smaller and intermediate-sized groups.
- Table IV Part 1 begins the regression analysis, controlling for log sales, age, and industry effects while examining Tobin's q and ROA.
- The coefficients on firm age are negative, consistent with younger firms having higher valuation ratios and profitability in this sample.

Detailed interpretation: Table III is the clearest visual/statistical version of the paper’s story. The average group affiliate is not the right object. The key question is which type of group the firm belongs to. Table IV translates this pattern into regression form and separates group membership, group diversification, group size, and firm controls.

Economic message: Group affiliation is valuable only when groups are sufficiently broad to provide institutional services at scale. At intermediate diversification, the costs of complexity and agency may dominate.

Table III
Tobin's q and ROA by Group Diversification and Group Size Categories

The table reports unadjusted means, unadjusted medians, and test statistics for differences of means and medians for the Tobin's q and ROA performance measures by categories of group size and group diversification. Tobin's q is approximated by (market value of equity + book value of preferred stock + book value of debt)/(book value of assets). ROA is calculated as (net income + (1 - tax rate) * interest)/(total assets), with a tax rate estimated for each firm. The diversification buckets are based on our simplest measure of diversification, the number of two-digit SIC-equivalent industries represented in the group. These are constructed by first creating the equivalent of an SIC system for Indian publicly traded firms. From the CMIE (Center for Monitoring the Indian Economy), our primary data source, we assign a two-digit SIC code to indicate each firm's primary area of activity based on descriptions of its product range. The least diversified category refers to groups in 1-4 industries, the intermediate diversified category refers to groups in 5-7 industries, and the most diversified category to groups in >7 industries. The group size buckets are based on our simplest measure of group size, the number of companies in the group. The smallest category corresponds to 1-7 firms, the intermediate-sized category to 8-17 firms, and the largest category to more than 17 firms. The number of groups column reports the number of groups across which the firms in the particular group diversification or group size category are distributed. The percentage of sample assets is the percentage of total assets in the sample accounted for by the firms in the particular group size or group diversification category, or in the group versus nongroup categories. All significance levels refer to comparisons of group affiliates of different group diversification and group size categories to nongroup firms. Two-tailed t-tests are used for the pairwise comparisons of means, and Wilcoxon signed-rank tests are used for the pairwise comparisons of medians.

Affiliation	Number of Firms	Number of Groups	Percentage of Sample Assets	Mean Tobin's q	Median Tobin's q	Mean ROA	Median ROA
Panel A: By Group Diversification Buckets (number of industries)							
Full sample	1309		100.00%	1.392	1.099	0.0853	0.0900
Nongroup	654		21.31%	1.382	1.066	0.0838	0.0890
All groups	655	272	78.68%	1.401	1.139**	0.0868	0.0915
Least diversified	469	241	45.86%	1.405	1.132*	0.0897	0.0949*
Intermediate diversified	123	24	18.04%	1.255**	1.070	0.0724*	0.0770**
Most diversified	63	7	14.77%	1.655**	1.393***	0.0937	0.0960
Panel B: By Group Size Buckets (number of companies)							
Full sample	1309		100.00%	1.392	1.099	0.0853	0.0900
Nongroup	654		21.31%	1.382	1.066	0.0838	0.0890
All groups	655	272	78.68%	1.401	1.139**	0.0868	0.0915
Smallest	507	253	49.70%	1.409	1.124**	0.0872	0.0920
Intermediate sized	100	15	16.25%	1.272	1.153	0.0835	0.0860
Largest	48	4	12.79%	1.590	1.283**	0.0894	0.0945

*** ** * denote significant differences from nongroup estimates at the 1 %, 5, and 10 percent levels, respectively.

Table III: Tobin's q and ROA by Group Diversification and Group Size Categories

Table IV
The Effects of Group Diversification and Group Size

This table reports OLS regressions examining the incremental effect of group membership on firm performance. Panel A uses an approximation to Tobin's q as a dependent variable. This is defined as (market value of equity + book value of preferred stock + book value of debt)/(book value of assets). Panel B uses ROA, defined as (net income + (1 - tax rate) * interest)/(total assets), where a tax rate is estimated for each firm. The sample consists of all Indian private sector firms publicly traded on the Bombay Stock Exchange in 1993. Data are obtained from the Centre for Monitoring the Indian Economy (CMIE). The standard errors reported in parentheses allow for correlation in the error terms among all members of a particular group, and correct for heteroskedasticity, following Moulton (1986, 1990). The independent variables are a set of firm-specific attributes and industry dummies and group attributes. The group dummy measures membership in a diversified group. The number of industries is used as a measure of group diversification. Number of firms is used as a measure of group size. A finer categorization of group dummies is used to capture the effects of membership in groups with different degrees of diversification. The least diversified group dummy captures membership in groups in 1-4 industries, the intermediate diversified group dummy captures membership in groups in 5-7 industries, and the most diversified group dummy captures membership in groups in >7 industries.

Variable	Panel A: Tobin's q			Panel B: ROA		
	Coefficient			Coefficient		
	(i)	(ii)	(iii)	(i)	(ii)	(iii)
Constant	1.2859*** (0.0778)	1.2863*** (0.0797)	1.3632*** (0.0836)	0.0600*** (0.0060)	0.0591*** (0.0066)	0.0682*** (0.0062)
Log of sales	0.0208 (0.0218)	0.0210 (0.0219)	0.0249 (0.0225)	0.0170*** (0.0018)	0.0170*** (0.0018)	0.0167*** (0.0018)
Age	-0.0051*** (0.0005)	-0.0050*** (0.0005)	-0.0053*** (0.0005)	-0.0008*** (0.0007)	-0.0007*** (0.0007)	-0.0008*** (0.0007)

Table IV: Effects of Group Diversification and Group Size, Part 1

7.3 Table IV: The Effects of Group Diversification and Group Size, Part 2 + Table A1: The Institutional Context in India

Read:

- Table IV Part 2 shows the key coefficients on group dummy variables, diversification, diversification squared, and group size.
- The group dummy alone is negative for ROA, suggesting that average group affiliation can be costly.
- The least-diversified and intermediate-diversified group dummies are negative in the ROA specification, while the most-diversified dummy is not significantly negative.
- The quadratic terms imply that marginal diversification initially reduces performance but eventually increases performance after a threshold.
- Group size is generally positive but statistically weak, indicating that industry breadth matters more than the number of firms.
- Table A1 compares the United States, Japan, and India across finance, labor, product markets, regulation, enforcement, and corporate scope.
- Table A1 summarizes the institutional-void logic: India has underdeveloped capital markets, weak monitoring, scarce management talent, limited product-market enforcement, high regulation, and unpredictable enforcement.

Detailed interpretation: The two tables are best read together. Table IV documents the empirical U-shape. Table A1 gives the institutional interpretation: large groups may be valuable precisely because India lacked well-functioning external institutions. The regressions tell us there is a pattern; the appendix table explains why the pattern is plausible in an emerging-market context.

Economic message: The most diversified groups may be large enough to build internal substitutes for missing institutions. Small and medium groups may incur agency costs without achieving enough scale to solve external market failures.

Appendix

The institutional context in which firms operate in India differs significantly from the context in advanced economies. The following table shows how the institutional context in India differs from that in the United States and Japan, two economies studied extensively by financial economists.

Table A1
The Institutional Context in India

Institutional feature	United States	Japan	India
Market for finance	Equity focused; managers monitored by disclosure rules and the market for corporate control	Bank focused; monitoring by interlocking investments and directors	Underdeveloped, illiquid equity markets and nationalized banks; weak monitoring by bureaucrats
Market for labor	Business schools and consulting firms offer talent; certified skills and welfare benefits enhance mobility	Few business schools; training takes place smoothly within firms, firm-specific development of talent	Few business schools; little training; management talent is scarce, workers inflexible
Market for products	Reliable enforcement of liability laws and efficient information dissemination, activist consumers	Reliable enforcement of liability laws and efficient information dissemination, less activist consumers	Limited enforcement of liability laws, little information dissemination, no activist consumers
Governmental regulation	Low, relatively corruption free	Moderate, relatively corruption free	High, corruption is common
Enforcement mechanisms	Predictable	Predictable	Unpredictable
Corporate scope	Focused; diversified groups have disadvantages	Groups may have some advantages	Groups may have many advantages

Source: Whv focused strategies may be wrong for emerging markets. Khanna and Palanu (1997).

Group dummy	0.0055 (0.0644)			-0.0126*** (0.0047)	
Least diversified group dummy		0.0062 (0.0674)			-0.0102** (0.0051)
Intermediate diversified group dummy		-0.1491 (0.1010)			-0.0275*** (0.0086)
Most diversified group dummy		0.2941*** (0.1104)			-0.0052 (0.0066)
Number of industries			-0.0722** (0.0338)		-0.0089*** (0.0027)
Number of industries squared			0.0060** (0.0028)		0.0006*** (0.0002)
Number of firms			0.0070 (0.0088)		0.0008 (0.0005)
Industry dummies					
Number of observations	1308	1308	1308	1303	1303
F-statistic	4.58***	5.0100***	7.4800***	7.0000***	6.5400***
R ²	0.0502	0.0560	0.0550	0.0913	0.0953
Root MSE	1.0005	1.0072	1.0080	0.0780	0.0749

***, **, and * indicate significance at the 1, 5, and 10 percent levels, respectively.

Table IV: Effects of Group Diversification and Group Size, Part 2

Table A1: The Institutional Context in India

8 Detailed result map

8.1 Result 1: No simple average group premium

- Group affiliates are bigger, older, and more institutionally owned than nongroup firms.
- Average performance differences between all group affiliates and all nongroup firms are modest.
- The average comparison masks large heterogeneity across group types.

8.2 Result 2: Nonmonotonic group-diversification effect

- Performance initially declines with group diversification.
- Performance improves once group diversification becomes high enough.
- The most diversified affiliates outperform on Tobin's q and do not underperform on ROA.

8.3 Result 3: Group size alone is not the main explanation

- Group size is sometimes positive but generally less robust than diversification breadth.
- The paper's interpretation depends on scope and scale of internal intermediation, not mere number of affiliates.

8.4 Result 4: Mechanism evidence favors institutional intermediation

- Group affiliates have better access to international capital markets.
- Insider ownership does not uniquely explain group-affiliate performance.
- Internal capital markets do not appear to be a major differentiating mechanism for ongoing investment.

8.5 Result 5: Institutional context matters

- The Indian setting is characterized by underdeveloped financial markets, scarce management talent, weak product-market institutions, high regulation, and unpredictable enforcement.
- Large diversified groups may create value by substituting for these missing institutions.
- The paper’s findings should not be directly mapped onto U.S. conglomerates because the institutions and organizational form differ.

9 Short summary

- Khanna and Palepu study whether Indian business groups create or destroy value.
- The sample contains 1,309 Bombay Stock Exchange firms in 1993, split almost equally between group affiliates and unaffiliated firms.
- Performance is measured using Tobin’s q and ROA.
- Diversification is measured using number of industries, entropy, concentric diversification, and Herfindahl concentration.
- The paper finds no simple unconditional group premium.
- It finds a U-shaped relation between performance and group diversification.
- Firms in intermediate-diversification groups underperform.
- Firms in the most diversified groups outperform on Tobin’s q and do not underperform on ROA.
- Group size alone does not explain the results.
- The most plausible interpretation is that large diversified groups can replicate missing institutions in emerging markets.
- The mechanism evidence is strongest for access to international capital markets.
- The evidence is weaker for internal capital markets and insider ownership as explanations.
- The paper emphasizes that institutional context and organizational structure change the welfare effects of diversification.

10 Conclusion

10.1 Core conclusion

The paper concludes that business-group affiliation in emerging markets has heterogeneous performance effects. Affiliation with a large majority of diversified Indian business groups is associated with lower performance relative to focused unaffiliated firms, but affiliates of the most diversified groups perform better than other firms on market-based performance measures. This pattern is very different from the standard U.S. conglomerate discount.

10.2 Economic intuition

The economic intuition is that emerging markets contain institutional voids. Capital markets, labor markets, product markets, information intermediaries, and contract enforcement mechanisms are incomplete. A sufficiently large business group can build internal substitutes for these institutions. It can use reputation to access suppliers and investors, use family and board networks to monitor managers, recruit and allocate managerial talent, interface with regulators, and help affiliates access foreign capital. But smaller or moderately diversified groups may not have enough scale to build such institutions and may still suffer from the agency costs of group affiliation.

10.3 Why the result is incremental

The paper changes how researchers think about diversified organizations. In developed economies, diversification is usually evaluated through the lens of agency costs and inefficient internal capital markets. Khanna and Palepu show that in emerging markets, the same organizational structure can be a second-best institutional substitute. Thus, the value of diversification is not universal; it depends on the institutional environment.

10.4 Policy implication

The findings imply that policies aimed at dismantling business groups or forcing narrow corporate focus may be misguided in emerging markets if alternative market institutions are not yet developed. A better policy objective is to develop external capital, labor, product, and governance institutions so that firms no longer need to rely on groups to perform these functions.

10.5 Limitations

The evidence is cross-sectional and does not fully solve endogeneity of group membership. The authors cannot rule out that the largest groups are run by unusually capable families or have superior political connections. The analysis is also based on one year, 1993, and one country, India. Therefore, the findings should be interpreted as strong evidence of an institutional-context mechanism, not as definitive causal proof of the effect of group membership.

10.6 Takeaway

The paper's central takeaway is that business groups can be either value destroying or value creating depending on scale, scope, and institutional environment. In India, the most diversified groups appear to add value because they are able to internalize market-supporting functions that are missing externally. This makes the paper a foundational contribution to the study of emerging-market business groups and the institution-based view of corporate strategy.